Lesson 39: MDP Model and Policies

AI/ML Mathematics

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1 Motivation and Overview

Consider an autonomous robot navigating a grid-world warehouse. At each time step the robot observes its current cell, selects a movement action (north, south, east, west, or stay), and receives a reward that depends on the cell reached – perhaps $+10$ for reaching the charging station, -1 for each movement to encourage efficiency, and -100 for colliding with an obstacle. The warehouse layout, the probabilistic effects of actions (wheels may slip on a wet floor), and the reward structure together define a sequential decision problem. How should we formalize this setting so that we can reason mathematically about optimal behavior? What class of decision rules should we consider, and how do we measure the long-run performance of a particular strategy?

The Markov decision process (MDP) provides the canonical mathematical framework for sequential decision-making under uncertainty. The tools developed in this lesson – the formal MDP tuple, policy classes, value functions, and the Bellman expectation equations – form the foundation upon which all of reinforcement learning is built.

Why MDPs Matter for RL/ML

- **Tabular RL algorithms**: Q-learning, SARSA, and Monte Carlo methods all operate within the MDP framework; their convergence proofs require the formal definitions introduced here.
- **Policy gradient methods**: REINFORCE and actor-critic algorithms optimize over parameterized policy classes, but the underlying objective (maximizing the value function) is defined through the MDP model.
- **Deep RL**: DQN, PPO, and SAC extend tabular methods using function approximation, yet their theoretical guarantees ultimately rely on MDP structure and Bellman equations.

Prerequisites. Probability spaces, conditional probability, and conditional expectation including the tower property (Lessons 9–12). Geometric series and absolute convergence (Lesson 1). Basic linear algebra: vectors and matrices (Lesson 5). Contraction mappings on complete metric spaces (Lesson 4, Lesson 15).

Learning objectives. After completing this lesson, the student will be able to:

- *Define* a Markov decision process and each of its components formally.
- *Classify* policies as history-dependent, Markov, stationary, deterministic, or stochastic, and *explain* the relationships among these classes.
- *Derive* the Bellman expectation equations for both the state-value function V^π and the action-value function Q^π.
- *Compute* value functions for small MDPs by direct solution of the Bellman equations.
- *State* the existence theorem for deterministic stationary optimal policies in finite MDPs.
- *Prove* that the return is well-defined and bounded under the discounted criterion.

2 References

- Puterman [2014, Chapters 2–3] – Definitive treatment of MDP formulation, policy classes, and optimality criteria.

- Bertsekas [2012, Chapter 1] – Concise development of dynamic programming foundations with emphasis on infinite-horizon discounted problems.
- Sutton and Barto [2018, Chapters 3–4] – Accessible introduction to MDPs, value functions, and Bellman equations from the RL perspective.
- Howard [1960] – Howard's seminal monograph introducing policy iteration and the MDP framework.
- Blackwell [1962] – Foundational paper on discounted dynamic programming and optimality.

3 The Markov Decision Process Model

Definition 3.1 (Markov Decision Process)**.** A *Markov decision process* (MDP) is a tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$ where:

1. $\mathcal{S}$ is a finite set of *states*, with $|\mathcal{S}| = S$.
2. $\mathcal{A}$ is a finite set of *actions*, with $|\mathcal{A}| = A$.
3. $P : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to [0, 1]$ is the *transition kernel*, where $P(s' \mid s, a) = \mathbb{P}(S_{t+1} = s' \mid S_t = s, A_t = a)$ satisfies $\sum_{s' \in \mathcal{S}} P(s' \mid s, a) = 1$ for every $(s, a) \in \mathcal{S} \times \mathcal{A}$.
4. $R : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ is the (deterministic) *reward function*, where $R(s, a)$ is the immediate reward received upon taking action a in state s.
5. $\gamma \in [0, 1)$ is the *discount factor*.

Remark 3.2. Several equivalent reward formulations appear in the literature: $R(s, a, s')$, $R(s)$, or random rewards with $\mathbb{E}[R_t \mid S_t = s, A_t = a] = R(s, a)$. In the finite-state, finite-action case with bounded rewards, these formulations are interchangeable up to a redefinition of the reward function. We adopt $R(s, a)$ throughout for notational simplicity. When using $R(s, a, s')$, the expected reward is $\bar{R}(s, a) = \sum_{s'} P(s' \mid s, a) R(s, a, s')$.

The key structural property of MDPs is the *Markov property*: the future evolution depends on the past only through the current state and action.

Definition 3.3 (Markov Property)**.** A stochastic process $(S_t)_{t \geq 0}$ controlled by actions $(A_t)_{t \geq 0}$ satisfies the *Markov property* if for all $t \geq 0$ and all histories $h_t = (s_0, a_0, s_1, a_1, \ldots, s_t, a_t)$,

$$\mathbb{P}(S_{t+1} = s' \mid S_0 = s_0, A_0 = a_0, \ldots, S_t = s_t, A_t = a_t) = \mathbb{P}(S_{t+1} = s' \mid S_t = s_t, A_t = a_t) = P(s' \mid s_t, a_t).$$

Example 3.4 (Grid World MDP)**.** Consider a 3×3 grid world with states $\mathcal{S} = \{(i, j) : i, j \in \{1, 2, 3\}\}$ and actions $\mathcal{A} = \{\mathrm{N}, \mathrm{S}, \mathrm{E}, \mathrm{W}\}$. The agent starts at $(1, 1)$ and aims to reach the goal at $(3, 3)$. The transition kernel is deterministic: $P((i + 1, j) \mid (i, j), \mathrm{N}) = 1$ if $i < 3$, and $P((i, j) \mid (i, j), \mathrm{N}) = 1$ if $i = 3$ (wall bounce), with analogous rules for the other directions. The reward is $R(s, a) = +1$ if the transition leads to $(3, 3)$, and $R(s, a) = 0$ otherwise. With $\gamma = 0.9$, this defines a complete MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, 0.9)$ with $S = 9$ states and $A = 4$ actions.

4 Policies

A *policy* specifies how the agent selects actions. We define a hierarchy of increasingly restrictive policy classes.

Definition 4.1 (History)**.** A *history* at time t is the sequence $h_t = (s_0, a_0, r_0, s_1, a_1, r_1, \ldots, s_{t-1}, a_{t-1}, r_{t-1}, s_t)$ recording all states visited, actions taken, and rewards received up to time t. Let $\mathcal{H}_t$ denote the set of all possible histories at time t, and $\mathcal{H} = \bigcup_{t=0}^{\infty} \mathcal{H}_t$.

Definition 4.2 (History-Dependent Policy)**.** A *history-dependent randomized policy* is a sequence $\pi = (\pi_0, \pi_1, \pi_2, \ldots)$ where each $\pi_t : \mathcal{H}_t \to \Delta(\mathcal{A})$ maps histories to probability distributions over actions. Here $\Delta(\mathcal{A}) = \{p \in \mathbb{R}^{|\mathcal{A}|} : p(a) \geq 0,\ \sum_a p(a) = 1\}$ denotes the probability simplex over $\mathcal{A}$. We write $\pi_t(a \mid h_t)$ for the probability of selecting action a given history h_t.

Definition 4.3 (Markov Policy)**.** A *Markov randomized policy* is a sequence $\pi = (\pi_0, \pi_1, \pi_2, \ldots)$ where each $\pi_t : \mathcal{S} \to \Delta(\mathcal{A})$ depends only on the current state (not the full history). That is, $\pi_t(a \mid h_t) = \pi_t(a \mid s_t)$ for all h_t ending in s_t.

Definition 4.4 (Stationary Policy)**.** A *stationary randomized policy* is a Markov policy that is time-independent: $\pi_t = \pi_0$ for all $t \geq 0$. We identify such a policy with a single mapping $\pi : \mathcal{S} \to \Delta(\mathcal{A})$ and write $\pi(a \mid s)$ for the probability of action a in state s.

Definition 4.5 (Deterministic Policy)**.** A *deterministic stationary policy* is a mapping $\pi : \mathcal{S} \to \mathcal{A}$ assigning a single action to each state. Equivalently, $\pi(a \mid s) = \mathbf{1}\{a = \pi(s)\}$. We denote the set of all deterministic stationary policies by Π^{SD} and the set of all stationary randomized policies by Π^{SR}.

Remark 4.6. The policy classes form a strict hierarchy:

$$\Pi^{\mathrm{SD}} \subset \Pi^{\mathrm{SR}} \subset \Pi^{\mathrm{MR}} \subset \Pi^{\mathrm{HR}},$$

where Π^{MR} and Π^{HR} denote the Markov randomized and history-dependent randomized policy classes, respectively. For the discounted infinite-horizon criterion, the optimal policy can always be found in Π^{SD}, so the larger classes offer no advantage (Theorem 7.3).

Stochastic Policies Are Not Redundant

Although deterministic policies suffice for optimality in standard MDPs, stochastic policies play an essential role in several settings: partially observable MDPs (POMDPs), multi-agent games (where mixed strategies are necessary for Nash equilibria), constrained MDPs (where randomization achieves the constraint boundary), and policy gradient methods (where stochastic policies enable gradient estimation).

Example 4.7 (Inventory Management MDP)**.** A retailer manages inventory of a single product. The state $s \in \mathcal{S} = \{0, 1, \ldots, M\}$ is the current stock level, and the action $a \in \mathcal{A} = \{0, 1, \ldots, M\}$ is the number of units ordered, subject to $s + a \leq M$ (warehouse capacity). Demand D is random with known distribution p_D. The transition kernel is $P(s' \mid s, a) = \mathbb{P}(\max(s + a - D, 0) = s')$. The reward is $R(s, a) = p \cdot \mathbb{E}[\min(s + a, D)] - c \cdot a - h \cdot \mathbb{E}[\max(s + a - D, 0)]$, where p is the unit selling price, c is the unit ordering cost, and h is the per-unit holding cost. A stationary deterministic policy $\pi(s) = a(s)$ specifies an order quantity for each stock level.

5 Return and Value Functions

Definition 5.1 (Discounted Return)**.** The *discounted return* from time t onward is the random variable

$$G_t = \sum_{k=0}^{\infty} \gamma^k R(S_{t+k}, A_{t+k}).$$

The first key result establishes that this infinite sum is well-defined.

Proposition 5.2 (The Return Is Well-Defined and Bounded)**.** *Let $R_{\max} = \max_{s,a} |R(s, a)|$. For any $\gamma \in [0, 1)$ and any policy π, the return G_t satisfies:*

1. *The series $\sum_{k=0}^{\infty} \gamma^k R(S_{t+k}, A_{t+k})$ converges absolutely almost surely.*

2. $|G_t| \leq R_{\max}/(1-\gamma)$ *almost surely.*
3. $\mathbb{E}[|G_t|] \leq R_{\max}/(1-\gamma)$.

Proof. For each $k \geq 0$, $|R(S_{t+k}, A_{t+k})| \leq R_{\max}$, so

$$\sum_{k=0}^{\infty} \gamma^k |R(S_{t+k}, A_{t+k})| \leq R_{\max} \sum_{k=0}^{\infty} \gamma^k = \frac{R_{\max}}{1-\gamma} < \infty.$$

Absolute convergence implies convergence, establishing (1). The same bound gives (2). Taking expectations and applying the monotone convergence theorem yields (3). □

Remark 5.3. The return satisfies the recursive relationship $G_t = R(S_t, A_t) + \gamma G_{t+1}$, which is the foundation of temporal-difference methods and the Bellman equations derived below.

Definition 5.4 (State-Value Function)**.** The *state-value function* of policy π is

$$V^\pi(s) = \mathbb{E}_\pi[\, G_t \mid S_t = s \,] = \mathbb{E}_\pi\left[\sum_{k=0}^{\infty} \gamma^k R(S_{t+k}, A_{t+k}) \;\middle|\; S_t = s \right],$$

where $\mathbb{E}_\pi$ denotes expectation under the probability measure induced by policy π and transition kernel P.

Definition 5.5 (Action-Value Function)**.** The *action-value function* of policy π is

$$Q^\pi(s,a) = \mathbb{E}_\pi[\, G_t \mid S_t = s,\ A_t = a \,].$$

Definition 5.6 (Advantage Function)**.** The *advantage function* of policy π is

$$A^\pi(s,a) = Q^\pi(s,a) - V^\pi(s).$$

It measures the relative advantage of taking action a in state s compared to acting according to π.

Proposition 5.7 (Value-Action Value Relationship)**.** *For any stationary policy π and state $s \in \mathcal{S}$,*

$$V^\pi(s) = \sum_{a \in \mathcal{A}} \pi(a \mid s)\, Q^\pi(s,a).$$

Consequently, $\sum_{a \in \mathcal{A}} \pi(a \mid s) A^\pi(s,a) = 0$ *for all* s.

Proof. By the law of total expectation (tower property),

$$V^\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s] = \sum_{a \in \mathcal{A}} \pi(a \mid s)\, \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a] = \sum_{a \in \mathcal{A}} \pi(a \mid s)\, Q^\pi(s,a).$$

The advantage identity follows immediately: $\sum_a \pi(a \mid s) A^\pi(s,a) = \sum_a \pi(a \mid s) Q^\pi(s,a) - V^\pi(s) = V^\pi(s) - V^\pi(s) = 0$. □

Advantage Functions in Policy Gradients

The advantage function $A^\pi(s,a)$ plays a central role in variance reduction for policy gradient methods. The policy gradient theorem (Lesson on Policy Gradients) shows that the gradient of the expected return can be written as $\nabla_\theta J(\theta) = \mathbb{E}[\nabla_\theta \log \pi_\theta(A_t \mid S_t) A^{\pi_\theta}(S_t, A_t)]$. Since $\sum_a \pi(a \mid s) A^\pi(s,a) = 0$, using the advantage as a baseline does not bias the gradient estimate but substantially reduces variance.

6 Bellman Expectation Equations

The Bellman expectation equations express the value of a state recursively in terms of the values of successor states. They are the most fundamental equations in the theory of MDPs.

Theorem 6.1 (Bellman Expectation Equation for V^π). *For any stationary policy π and any state $s \in \mathcal{S}$,*

$$V^\pi(s) = \sum_{a\in\mathcal{A}} \pi(a \mid s) \left[R(s,a) + \gamma \sum_{s'\in\mathcal{S}} P(s' \mid s,a)\, V^\pi(s') \right]. \tag{1}$$

Proof. Starting from the recursive decomposition of the return, $G_t = R(S_t, A_t) + \gamma G_{t+1}$, and conditioning on $S_t = s$:

$$\begin{aligned} V^\pi(s) &= \mathbb{E}_\pi[G_t \mid S_t = s] \\ &= \mathbb{E}_\pi[R(S_t, A_t) + \gamma G_{t+1} \mid S_t = s] \\ &= \sum_{a\in\mathcal{A}} \pi(a \mid s) \mathbb{E}_\pi[R(s,a) + \gamma G_{t+1} \mid S_t = s, A_t = a] \\ &= \sum_{a\in\mathcal{A}} \pi(a \mid s) \left[R(s,a) + \gamma \mathbb{E}_\pi[G_{t+1} \mid S_t = s, A_t = a] \right]. \end{aligned}$$

By the Markov property and stationarity of π,

$$\mathbb{E}_\pi[G_{t+1} \mid S_t = s, A_t = a] = \sum_{s'\in\mathcal{S}} P(s' \mid s,a)\, \mathbb{E}_\pi[G_{t+1} \mid S_{t+1} = s'] = \sum_{s'\in\mathcal{S}} P(s' \mid s,a)\, V^\pi(s').$$

The last equality uses stationarity: the value function V^π is time-independent under a stationary policy. Substituting yields (1). □

Theorem 6.2 (Bellman Expectation Equation for Q^π). *For any stationary policy π and any $(s,a) \in \mathcal{S} \times \mathcal{A}$,*

$$Q^\pi(s,a) = R(s,a) + \gamma \sum_{s'\in\mathcal{S}} P(s' \mid s,a) \sum_{a'\in\mathcal{A}} \pi(a' \mid s')\, Q^\pi(s',a'). \tag{2}$$

Proof. Decompose the return as $G_t = R(S_t, A_t) + \gamma G_{t+1}$ and condition on $(S_t, A_t) = (s,a)$:

$$\begin{aligned} Q^\pi(s,a) &= \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a] \\ &= R(s,a) + \gamma\, \mathbb{E}_\pi[G_{t+1} \mid S_t = s, A_t = a] \\ &= R(s,a) + \gamma \sum_{s'} P(s' \mid s,a)\, \mathbb{E}_\pi[G_{t+1} \mid S_{t+1} = s'] \\ &= R(s,a) + \gamma \sum_{s'} P(s' \mid s,a)\, V^\pi(s'). \end{aligned}$$

Applying Proposition 5.7 to replace $V^\pi(s')$ gives (2). □

Bellman Expectation Equations – Matrix Form

For a stationary policy π, define the $S \times S$ transition matrix P^π with entries $P^\pi_{ss'} = \sum_a \pi(a \mid s) P(s' \mid s,a)$ and the reward vector $r^\pi \in \mathbb{R}^S$ with entries $r^\pi_s = \sum_a \pi(a \mid s) R(s,a)$. Then the Bellman equation (1) becomes $v^\pi = r^\pi + \gamma P^\pi v^\pi$, which has the unique solution

$$v^\pi = (I - \gamma P^\pi)^{-1} r^\pi.$$

The matrix $I - \gamma P^\pi$ is invertible because all eigenvalues of γP^π have modulus at most $\gamma < 1$ (since P^π is a stochastic matrix).

Example 6.3 (Two-State MDP)**.** Consider an MDP with $\mathcal{S} = \{s_1, s_2\}$, $\mathcal{A} = \{a_1, a_2\}$, $\gamma = 0.5$, and:

s	a	$R(s,a)$	$P(s_1 \mid s,a)$	$P(s_2 \mid s,a)$
s_1	a_1	5	0.5	0.5
s_1	a_2	10	0	1
s_2	a_1	-1	0.8	0.2
s_2	a_2	2	0.3	0.7

Under the deterministic policy $\pi(s_1) = a_1$, $\pi(s_2) = a_1$, the Bellman equation in matrix form is $v = r + 0.5P^\pi v$ where $r = (5, -1)^\top$ and

$$P^\pi = \begin{pmatrix} 0.5 & 0.5 \\ 0.8 & 0.2 \end{pmatrix}.$$

Solving $(I - 0.5P^\pi)v = r$:

$$\begin{pmatrix} 0.75 & -0.25 \\ -0.4 & 0.9 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 5 \\ -1 \end{pmatrix},$$

which gives $v_1 = \frac{0.9\cdot 5 + 0.25\cdot(-1)}{0.75\cdot 0.9 - 0.25\cdot 0.4} = \frac{4.25}{0.575} \approx 7.39$ and $v_2 = \frac{0.75\cdot(-1)+0.4\cdot 5}{0.575} = \frac{1.25}{0.575} \approx 2.17$.

7 Optimal Policies and Value Functions

Definition 7.1 (Optimal Value Functions)**.** The *optimal state-value function* is $V^*(s) = \sup_{\pi \in \Pi^{\text{HR}}} V^\pi(s)$ for all $s \in \mathcal{S}$. The *optimal action-value function* is $Q^*(s,a) = \sup_{\pi \in \Pi^{\text{HR}}} Q^\pi(s,a)$ for all $(s,a) \in \mathcal{S} \times \mathcal{A}$.

Definition 7.2 (Optimal Policy)**.** A policy π^* is *optimal* if $V^{\pi^*}(s) = V^*(s)$ for all $s \in \mathcal{S}$.

Theorem 7.3 (Existence of Deterministic Stationary Optimal Policy)**.** *For any finite MDP $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$ with $\gamma \in [0,1)$, there exists a deterministic stationary policy $\pi^* \in \Pi^{\text{SD}}$ that is optimal. That is, $V^{\pi^*}(s) = V^*(s)$ for all $s \in \mathcal{S}$.*

Proof. The proof relies on the Bellman optimality equations, which are developed in detail in Lesson 41. We outline the argument here; the complete proof appears there.

Since $\mathcal{S}$ and $\mathcal{A}$ are finite, there are finitely many deterministic stationary policies: $|\Pi^{\text{SD}}| = A^S$. Each deterministic stationary policy $d \in \Pi^{\text{SD}}$ has a well-defined value function V^d, computable via the matrix formula $V^d = (I - \gamma P^d)^{-1} r^d$. Define $\hat{V}(s) = \max_{d \in \Pi^{\text{SD}}} V^d(s)$. It can be shown (Lesson 41, using the Bellman optimality operator as a contraction) that there exists a single deterministic policy d^* achieving $V^{d^*}(s) = \hat{V}(s)$ for all s simultaneously, and that $\hat{V} = V^*$, i.e., no history-dependent or randomized policy can improve upon d^*. □

Why Deterministic Optimal Policies Exist

Theorem 7.3 justifies the design of algorithms like Q-learning that search for deterministic policies. The greedy action $\arg\max_a Q(s,a)$ is always sufficient – there is no need to maintain or learn a stochastic policy for optimality in standard MDPs. This is why DQN and its variants output a single best action per state, rather than a probability distribution over actions.

Example 7.4 (Optimal Policy for Two-State MDP)**.** Returning to Example 6.3, we can evaluate all $|\Pi^{\mathrm{SD}}| = 2^2 = 4$ deterministic policies. Solving the Bellman equations for each:

$\pi(s_1)$	$\pi(s_2)$	$V^\pi(s_1)$	$V^\pi(s_2)$
a_1	a_1	7.39	2.17
a_1	a_2	8.50	5.00
a_2	a_1	11.30	3.48
a_2	a_2	14.00	6.00

The optimal policy is $\pi^*(s_1) = a_2$, $\pi^*(s_2) = a_2$, with $V^*(s_1) = 14$ and $V^*(s_2) = 6$. As guaranteed by Theorem 7.3, a single deterministic stationary policy achieves the maximum simultaneously at both states.

8 Exercises

Exercise 8.1 (MDP Formulation)**.** $(*)$
A taxi operates in a 5×5 grid city. The state includes the taxi's position, whether it is carrying a passenger, and the passenger's destination (one of 4 designated locations). The taxi can move in four cardinal directions, pick up, or drop off. Formally define this problem as an MDP: specify $\mathcal{S}$, $\mathcal{A}$, the transition kernel P, reward function R, and an appropriate discount factor γ.

Exercise 8.2 (Bellman Equation Solution)**.** $(*)$
Consider a three-state MDP with $\mathcal{S} = \{s_1, s_2, s_3\}$, $\mathcal{A} = \{a\}$ (a single action), $\gamma = 0.9$, rewards $R(s_1) = 1$, $R(s_2) = 2$, $R(s_3) = 0$, and transition matrix

$$P = \begin{pmatrix} 0.2 & 0.5 & 0.3 \\ 0.1 & 0.6 & 0.3 \\ 0.4 & 0.1 & 0.5 \end{pmatrix}.$$

Compute $V^\pi(s)$ for each state by solving the linear system $(I - \gamma P)v = r$.

Exercise 8.3 (Advantage Function Properties)**.** $(**)$
Let π be a stationary policy for a finite MDP.

(a) Prove that $\sum_a \pi(a \mid s) A^\pi(s, a) = 0$ for all s.

(b) Show that if π is a deterministic policy with $\pi(s) = a^*$, then $A^\pi(s, a^*) = 0$ and $A^\pi(s, a) = Q^\pi(s, a) - Q^\pi(s, a^*)$ for all a.

(c) Prove that a stationary policy π is optimal if and only if $A^\pi(s, a) \le 0$ for all (s, a).

Exercise 8.4 (Inventory MDP Analysis)**.** $(**)$
For the inventory MDP of Example 4.7 with $M = 2$, demand $\mathbb{P}(D = 0) = 0.3$, $\mathbb{P}(D = 1) = 0.5$, $\mathbb{P}(D = 2) = 0.2$, $p = 4$, $c = 2$, $h = 1$, and $\gamma = 0.95$:

(a) Write out the transition matrix $P(s' \mid s, a)$ for each feasible state-action pair.

(b) Compute the reward $R(s, a)$ for each feasible pair.

(c) Evaluate V^π for the policy "order up to 2" (i.e., $\pi(s) = 2 - s$).

Exercise 8.5 (Policy Hierarchy and Counterexample)**.** $(***)$
Theorem 7.3 states that deterministic stationary policies suffice for optimality in finite discounted MDPs. Construct an explicit example demonstrating that this fails for constrained MDPs: give a two-state, two-action MDP with a constraint on average cost where the optimal constrained policy is necessarily stochastic (i.e., no deterministic stationary policy is optimal among policies satisfying the constraint).

Hint: Consider a constraint of the form $\mathbb{E}_\pi[\sum_t \gamma^t c(S_t, A_t)] \le C$ with appropriately chosen costs c and threshold C.

9 Summary

- A Markov decision process $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$ is the canonical framework for sequential decision-making under uncertainty, defined by states, actions, transitions, rewards, and a discount factor.
- The Markov property ensures that the future depends on the past only through the current state and action, enabling recursive decompositions.
- Policies range from history-dependent randomized (most general) to deterministic stationary (most restrictive): $\Pi^{\mathrm{SD}} \subset \Pi^{\mathrm{SR}} \subset \Pi^{\mathrm{MR}} \subset \Pi^{\mathrm{HR}}$.
- The discounted return $G_t = \sum_{k=0}^{\infty} \gamma^k R(S_{t+k}, A_{t+k})$ is well-defined and bounded by $R_{\max}/(1-\gamma)$ whenever $\gamma < 1$.
- The state-value function $V^\pi(s)$ and action-value function $Q^\pi(s,a)$ are linked by $V^\pi(s) = \sum_a \pi(a \mid s) Q^\pi(s,a)$.
- The advantage function $A^\pi(s,a) = Q^\pi(s,a) - V^\pi(s)$ measures the relative merit of action a; its expected value under π is zero.
- The Bellman expectation equations express V^π and Q^π recursively and, in the finite case, reduce to a linear system with unique solution $v^\pi = (I - \gamma P^\pi)^{-1} r^\pi$.
- For finite MDPs with $\gamma < 1$, an optimal deterministic stationary policy always exists.

10 Further Reading

- **Lesson 40** extends the MDP framework to finite-horizon problems, develops backward induction, and proves the principle of optimality.
- **Lesson 41** derives the Bellman optimality equations, introduces the Bellman operators $\mathcal{T}^\pi$ and $\mathcal{T}^*$, and proves the optimality of the greedy policy with respect to V^*.
- Puterman [2014, Chapters 4–6] provides an exhaustive treatment of optimality criteria, including average reward and total reward formulations beyond the discounted setting.
- Blackwell [1962] proves the existence of optimal stationary policies using contraction arguments; the original paper remains highly readable.
- Altman [1999] extends the MDP framework to constrained settings where stochastic policies are necessary for optimality, complementing Exercise 5 above.

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